

The impact of financial leverage on cost of capital and valuation of selected firms under the construction, infrastructure & allied services category of the Philippine Stock Exchange

**In Partial Fulfillment
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CHAPTER I

INTRODUCTION

1.1 Background of the Study

As defined by Odit (2008), "Financial leverage is the amount of debt used to finance a firm's assets and projects." Financial Leverage is referred to as the extent to which a firm relies on debt. When a firm uses more debt financing in its capital structure, it implies that there is a high degree of financial leverage.

Bhayani (2009) stated that the firm uses its owners' funds and outside financing to fund its assets, and effectively is the composition of the firm's capital structure. The total cost of capital was described as the aggregate costs of funds from specific sources. Financial managers sought to know the importance of financial leverage in maximizing shareholder's wealth. As stated by Ross, Westerfield, and Jordan (2003), minimizing the weighted average cost of capital would maximize the firm's value. The optimal capital structure is the ideal level of debt to equity of a firm, since this would enable the firm to maximize the firm value while minimizing the overall cost of capital. With that said, the researchers looked to examine if there was an impact by the degree of financial leverage to the cost of capital and market value of the firm. The researchers also aimed to find an optimal capital structure for the firms included in the study.

Studies in the area of capital structure and cost of capital have been carried out. However, most of them were based in developed countries such as USA, UK, etc. Meanwhile, there were also studies about capital structure conducted in developing

countries such as India. A study conducted by Sharma and Rao (1969) found that cost of capital was affected by debt apart from tax benefits. Several studies in India also indicated that profitability of a firm is positively related to its financial leverage. According to Bhayani (2009), despite the studies conducted in capital structure, none has delved into the impact of financial leverage to the cost of capital and valuation of firm. Thus, Bhayani focused his research on this area through his article, “Impact of Financial Leverage on Cost of Capital and Valuation of Firm: A Study of Indian Cement Industry.” Another study focused their research on the causal relationship of capital structure and cost of capital, as documented in their research entitled, “The Causal Relationship between Capital Structure and Cost of Capital: Evidence from ICT companies listed at NASDAQ” (Auon and Hesmati 2009).

The researchers have applied the researches done by Bhayani, Aoun and Heshmati to the selected companies listed under the Construction, Infrastructure & Allied Services classification of the Philippine Stock Exchange. Specifically, EEI Corporation, Holcim Philippines, Inc., Phinma Corporation, Republic Cement Corporation, Southeast Asia Cement Holdings, Inc., TKC Steel Corporation, and Vulcan Industrial & Mining

1.2 Statement of the Problem

The proponents of this paper stated the problem statement in the following manner:

What is the impact of financial leverage on cost of capital and market value of firm and what is the optimal capital structure of the selected firms under the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange?

1.3 Objectives of the Study

The objectives of this study are as follows:

1. To study the impact of financial leverage on the average cost of capital of selected firms under the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange for the period 2001-2009.
2. To analyze the impact of financial leverage on price-earning ratio and market value of selected firms under the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange for the period 2001-2009.
3. To observe if there is a correlation of financial leverage with cost of capital, price-earning ratio and market value of selected firms under the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange for the period 2001-2009.
4. To find the optimal capital structure for the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange using the cost of capital approach.

1.4 Statement of Hypotheses

Null Hypothesis (H_{o1}): There is no correlation between degree of financial leverage and cost of capital.

Alternative Hypothesis: (H_{a1}): There is a significant correlation between degree of financial leverage and cost of capital

Null Hypothesis (H_{o2}): There is no correlation between degree of financial leverage and the price earning ratio.

Alternative Hypothesis: (H_{a2}): There is a significant correlation between degree of financial leverage and the price earning ratio.

Null Hypothesis (H_{o3}): There is no correlation between degree of financial leverage and valuation of the firm.

Alternative Hypothesis: (H_{a3}): There is a significant correlation between degree of financial leverage and valuation of the firm.

Null Hypothesis (H_{o4}): Financial leverage does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a4}): Financial leverage statistically influences cost of capital.

Null Hypothesis (H_{o5}): Firm size does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a5}): Firm size statistically influences cost of capital.

Null Hypothesis (H_{o6}): The firm's systematic risk does not statistically influence cost of capital.

Alternative Hypothesis: (H_{a6}): The firm's systematic risk statistically influences cost of capital.

1.5 Assumptions

This study is based on the assumptions that the firms under the Construction, Infrastructure & Allied Services sub-sector are comparable with respect to their behavior towards capital structure, business activities, and risk-sensitivity. All the secondary data acquired from the BLOOMBERG are reliable and accurate.

1.6 Scope and Limitations

The scope of this paper is to study the impact of financial leverage on cost of capital and market value of selected firms under the classification Construction, Infrastructure & Allied Services of the Philippine Stock Exchange. The study also attempted to derive the optimal capital structure of the sub-sector from a regression model. The firms were chosen on the basis of availability of data.

There are many issues concerned with capital structure. However, for this paper, the researchers will limit the study on the impact of financial leverage on cost of capital and market value of firm. The measures used for valuing the firms, determining cost of capital and degree of financial leverage are limited to the availability of data from BLOOMBERG and the advice of the thesis advisor. The regression model used is limited to the significance of the individual variables to the dependent variable and the model provided by Aoun and Heshmati. Furthermore, the study will only cover the period 2001-2009, due to availability of data.

1.7 Significance of the Study

In real life, there is a trade-off between the tax benefits of debt and the risk of bankruptcy. Finding the right mix of debt and equity wherein the tax benefits of debt offsets its risk of bankruptcy generates the optimal capital structure of the firm. The optimal capital structure minimizes the cost of capital and maximizes the value of the firm. Therefore, financial managers have tried to achieve this ideal mix of capital

structure. Based on Bhayani (2009), in an empirical framework, financing decisions are not residuals of other managerial decisions, such as investment or strategic decisions. For Odit (2008), financial leverage can magnify gains or losses. With its proper use, financial leverage can lead to an increase in shareholder's wealth. Its misuse, however, can lead to risk of bankruptcy and loss of shareholder's value. With that said, the importance of financial leverage to financial managers cannot be stressed enough.

The impact of financial leverage on cost of capital and firm valuation would affect the capital structure decisions of firms. Management would now have a better understanding on the possible repercussions they may face with their use of financial leverage. Creditors would also benefit from this study as they would understand the impact of extending credit to firms based on their degree of financial leverage.

Such study has not yet been conducted in the Philippine setting. Therefore, the study would benefit similar firms, as they would gain better understanding on the effects of their capital structure decisions to their cost of capital and shareholder's wealth. There were also minimal researches done on the Construction, Infrastructure & Allied Services sub-sector of the Philippine Stock Exchange. The optimal capital structure generated for the sub-sector would benefit the companies that were included in it. We envisioned that our research would lead to more research done on this oft-overlooked sub-sector of the Philippine Stock Exchange. In the process of researching, the proponents of this paper gained significant knowledge on the subject matter, thereby supplementing their

knowledge in this specific finance topic. With that said, the researchers were able to apply the knowledge they acquired throughout their course program.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Previous Studies

This chapter discusses previous studies done on financial leverage, capital structure, cost of capital and optimal debt level. Particularly, this chapter aims to categorize the areas in capital structure practices. Areas in capital structure that is already established and still unidentified. It is also in concurrence with the objectives of the researchers to put emphasis on which principles are opposed and challenged in general literature. Aside from these, the construction of this chapter will help the proponents of the study to identify and find possible explanations to problems that will arise. The literature map can be referred to on Appendix A.

Modigliani and Miller (1958) were among the first ones to study about capital structure. In their work, it stated that capital structure has no influence with the firm's value under definite assumptions. In these assumptions, bankruptcy costs, costs of issuing securities and taxes were excluded but investment opportunities are present. Therefore, it was assumed that in the absence of a perfect capital market, firm's was primarily valued based on their financing preferences – the choice between debt and equity. The primary question was what would be the best combination of debt and equity in a particular industry. Basically, choosing one over the other would give certain benefits and

downfalls. Choosing the best combination of debt and equity is difficult to do because debt as itself has interest and interest would vary over time while equity has the capacity to appreciate or depreciate through time.

Specified by Milken (1958), when a company is involved in a technology-based industry such as those businesses that are associated with computers and electronics, volatility are natural, therefore, its balance sheet must be sound. There must be adequate capital and liquidity – the objectives of the optimal mix – in order for the company businesses to continue. Progressively, times are becoming difficult in the business setting. Modernization is increasing, and companies must deal with all the challenges in the business cycle as well as the global, financial, and economic setting. In the Construction, Infrastructure and Allied Services sector, companies do need to have modern technology to efficiently do their business processes. Volatility in the market prices of construction materials affects the financing of construction companies, so it is very important that they would have good capital structure. Needless to say, achieving the optimal mix when it comes to financing decisions is a strategy companies can exploit to have advantages in their industries.

As per Ross, Westerfieldm and Jordan (2003), optimal capital structure is finding the ideal mixture of debt and equity of a company. It is where value of the company maximizes and minimizes the overall cost of capital. The company's capital structure is irrelevant when it ignores taxes, financial distress costs, and any other imperfections. If one have considered the effect of corporate taxes, he would found out that capital structure matters. This conclusion was based on the fact that interest is tax deductible and

thus generates a valuable tax shield. Then, if one considered bankruptcy cost or financial distress. It stated that optimal capital structure exists when the net tax saving from an additional dollar in interest just equals the increase in expected bankruptcy costs.

Baker and Wurgler (2000) believed that there was no optimal capital structure. It is mainly based from collective result of previous efforts to time the equity market. Therefore, this timing is the relevant aspect that both the investors and the managers should look into in dealing with financial policies and decisions. With this in mind, the choice of managers on the type of financing has no use. Also, according to Herczeg (2004), despite the theoretical appeals, researchers in financial management have not found the optimal structure. These studies were patterned after the pioneers Modigliani and Miller (1958).

The Pecking Order Theory explains that companies prefer the use of internal to external financing and debt to equity in deciding for their investments. This theory presented a better experimental explanation of capital structures than the traditional tradeoff model (Shyam-Sunder and Myers, 1999). However, previous studies have already explained how the Pecking Order Theory was a better option against the Static Tradeoff Theory. On the other hand, as more and more people continued to look at in both theories' weaknesses, the researchers found more facts why these two could not be left unaccompanied by other models or theories. Frank and Goyal (2000) further examined the restrictions of the Pecking Order theory in which they made clear that equity is not in the bottom of the financial ladder. They found that particular

characteristics of companies affect to their financing decisions and not just merely the use of equity as a last alternative. In Myers' study (1984), he concluded that companies have their own rationales for financing real investment by using debt instead of issuing stock or other risky securities. But these do not stop the Pecking Order Theory from providing what its model suggests. Evidences still suggest that there are firms who accumulate their profits and losses in accordance with the Pecking Order Theory.

On how large capital expenditures and acquisitions of companies were financed was the aim of the study of Elsas, et al (2006). They examined companies from the United States during the period 1989-1999. It was evident from this study that capital expenditures and acquisitions were always followed by a financing decision. It was because firms assessed how to best back up their investment decisions. The results of the study were consistent with the Pecking Order theory in which internal financing is the first choice of backing up investments, next comes debt, and then equity.

De Meza and Webb (1987) provided the idea that asymmetric information played a vital role in knowing the level of investment and the possible project returns after venturing into such investment. The study analyzed the effects of asymmetric information on collective investment and on the financial structure of companies. The paper made use of a simple competitive model formulated by the researchers which showed that the inability of banks to discover features of projects may lead to more investment. The result was compared to Stiglitz and Weiss (1981) as well as other papers concerning the study made by Myers and Majluf (1984), which preferred the use of external financing using

debt rather than by equity in corporate financing decisions when they had the essential information which the investors do not have.

Fualkender and Petersen connected this information in their study. They figured out that the very little information small private companies have affected their credit constraints because of the high costs of collecting relevant information. Therefore, the diverse effects on companies may be seen in the capital markets when shocks occur to companies. They found out that asymmetric information, together with transaction costs, does not really affect the constraints of equity problems. Equity problems have benefits that offset the costs which why would choose this over debt.

The study of Jensen and Meckling (1976) determined that agency costs were presented in both equity and debt financing, but this can be reduced if optimal mixture was attained. Ross (1977) proposed that a manager of a company whose salary relies upon current and future company values will choose to signal the quality of the company through debt. In this study of Narayanan (1988), the use of debt was advantageous because it kept away substandard companies, which maintain the quality of companies in the market. Substandard companies have had a hard time in repaying its debts and therefore it would force them to borrow again or shutdown their business. Having a good competition in the market leads to a healthier economy. It is in this sense that every company in the market will be treated averagely, therefore allowing them to be valued at the same level. Results of the study concluded that if the company was undervalued, debt financing should be used by the financial managers. On the contrary, if the particular

company was overvalued, the company must use equity. According to the study conducted by Leite (1998), when high control benefits are apparent and it is coupled with more long-term debt, there may be a high level of debt.

In Faulkender and Petersen's (2005) study, leverage was measured as a company's debt-to-asset ratio with the ratio on book and market value. Companies were taken from 1986-2000 where financial companies, public sectors and those with less than \$1 million sales were not included. In this study on the Philippine sector, only those that were publicly listed on the Philippine Stock Exchange Construction, Infrastructure and Allied Services sector were included. Moreover, the study excluded financial companies and utilities due to the different treatments in the financial statements of financial institutions and regulated capital financing of utilities companies. Banks, for example, have very limited investments in net fixed assets; while on the other hand, construction companies invest much of their money to fixed assets. Utilities companies that are regulated do not have the ability to change prices compared to the construction companies.

The study conducted by Faulkender and Petersen (2005) regarding the source of capital and its effects on capital structure examines the correlation between the two. The study first considered the leverage of companies as a function of one's capital market access, and with the use of a regression equation, it examines the role of credit limitations and dissimilarity of public and private debt markets. Other than that approach, it also pursues an act of estimation or prediction on the variations of supply and which among the firms are able to access the public bond markets so as to distinguish those who can get a rating or not. To note, leverage is measured as a company's debt-to-asset ratio with

the debt ratio on both book value and market value basis. Sample companies were taken from Compustat from the period 1986-2000; all of which do not include the financial and public sector and those with less than \$1 million sales.

In the study conducted by Bhat (1980), he concluded that business risk, profitability, dividend pay-out and debt-service capacity are significant determinants of leverage. It shows that the capability of a firm to manage its cash flow would determine its leverage because basically it is how the firm would repay its debt. A firm's profitability would help them to borrow because it can show the lender their financial capability to repay. Paying of dividends to stockholders suggests that the company is profitable and the company's stock is getting valuable. Venkatesan (1983) studied the impact of sales and cash flow coverage. He concluded that cash flow coverage has a significant impact in determining a firm's leverage ratio because managing the cash flow right would lead to less risk transactions in the market. Sharma and Rao (1969) found out through the study they made on capital structure and cost of capital that cost of capital was affected by debt aside from tax advantages.

Scott (1977) argues that the issuance of a debt that is secured can increase the total value of a firm, even in the absence of taxes. The study shows that in taxless and frictionless markets where there is no chance that the firm would go into bankruptcy, changes in its debt-equity ratio will not fluctuate the total market value of its debt along with its equity. It is suggested that when companies are below of their current value, companies should prefer using debt financing over equity. In the study, he had certain

assumptions that led to his findings. In these assumptions, a securities market which is composed of a great number of buyers, sellers and issuers of financial instruments have a large impact in market prices where there is no floatation costs, transfer taxes, costs of bankruptcy and other transaction costs are completely isolatable. Choosing debt would be more advantageous for the undervalued companies because the use of debt financing would increase their company's total value. Having the optimal mix of debt and equity would increase the value of the company because it would maximize its resources.

In the study of Aoun and Heshmati (2006) that examines the relationship between international diversification, capital structure, and cost of capital, they conclude that there is an inverse relation between the cost of capital and leverage. By increasing leverage, the company's cost of capital decreases due to the fact that the cost of debt is typically lower than the cost of equity; consequently, higher levels of debt would decrease the weighted average of cost of capital. Another explanation could be that the unit cost of debt is declining in the level of debt.

On the other hand, in the paper by Bhayani (2009), he found that financial leverage has no impact; specifically there was no linear relationship, on cost of capital in the Indian cement industry. There was also no correlation between financial leverage and the total value of the companies in the Indian cement industry and therefore, it does not support the Modigliani and Miller approach that levered companies have higher valuation than un-levered companies. Lastly, the total valuation of a firm may increase through

various combination of three variables – cost of capital, financial leverage and price earnings ratio.

Due to the global financial crisis that happened from the end of 2007 to the start of 2009, most of the companies all over the world did not do well operationally and financially. Most of the big companies in the United States went bankrupt. In the Philippines, according to Tiongson (2010), most of the companies were affected but not hit hard by the global financial crisis. The Philippines did not experienced the recession but it faced the crisis with very little economic growth. The property sector is the least affected sector in the Philippines due to the continuous construction of condominiums and villages. As supplier of the property sector, the construction sector was also not affected by the global financial crisis. Remittances from Overseas Filipino Workers (OFWs) have continued to grow but at a slower rate. It is where the families of the OFWs get their money to be able to buy condominiums and houses.

As of the present, the economic health of the Philippines is good because of the booming businesses. Businessmen expect a more buoyant economy this year as business optimism hit a new all-time high for the next quarter after a slight dip in the current quarter due to the usual slowdown in business activity after the Christmas season. According to Bangko Sentral ng Pilipinas (BSP) director of the Department of Economic Statistics, the Business Expectations Survey for the first quarter of 2011 showed the confidence index reaching an all-time high of 59.4% in the next quarter due to government plans to undertake big infrastructure projects. Business confidence remained

highest in the construction sector. The survey showed that business outlook index on the macroeconomy for the next quarter of the construction sector hit a new all-time high of 65.2%. This means that the construction sector would be on the rise when it comes to industry growth.

2.2 Research Gap

This paper was based on a former study made by Bhayani, in 2009, entitled “Impact of Financial Leverage on Cost of Capital and Valuation of Firm: A Study of Indian Cement Industry.” In the aforementioned dissertation, Bhayani analyzed the effect of financial leverage on cost of capital and on valuation of firm in the cement industry. His findings showed that there is no relationship between financial leverage and cost of capital and there is no correlation between it and the valuation of firms.

It is essential for firms to find their optimal debt level in order to maximize their firm’s value. Studies of similar nature typically used the study of Modigliani and Miller as their basis, proving or disproving it. The case of Bhayani, it disproved the study made by Modigliani and Miller about the premise that financial leverage affects company’s value. Bhayani concluded that financial leverage had no correlation to the total value of Indian cement companies. Moreover, even if a lot of literature has been made about capital structure – choosing debt or equity, the study focused on the Philippine setting specifically the Construction, Infrastructure and Allied Services sub-sector in the publicly listed companies in the Philippine Stock Exchange. We only chose seven companies due

to the fact that they are ones that is highly tradable in the PSE. We assume that the seven companies represent the whole sub-sector.

CHAPTER III

FRAMEWORK OF THE STUDY

This chapter explained the different types of framework of this study. First, Theoretical Framework contained the underlying principles this study was based on. The theories to be presented were the groundwork and explanation of the event undertaken in the study. Second, Conceptual Framework dealt with the variables and their interrelationship between each other as they provided impact on the fundamentals of the study. Lastly, Operational Framework provided explanation of the model used and was the foundation process of the system. It also featured the different terminologies and how it was used specifically in this study.

3.1 Theoretical Framework

3.1.1 *Modigliani-Miller Theorem*

For many years, the M&M theorem had been associated with many different financial theorems; one of the well-known theorems would be regarding the capital structure and dividend policy. In the article written by Alan Levinsohn (2003) “Modigliani and Miller lives on”, the two propositions of M&M theorem had been discussed; the first proposition stated that whether or not the capital structure was levered, it had no effect on the value of the firm, by considering perfect capital market and tax neutrality. This proposition determined that real assets were the key determinant on the value of the firm where the optimal capital structure was 100% debt. As for the

second proposition, according to Modigliani and Miller in 1961, dividend policy was not important since a firm's investors would not be affected by its decision to pay all, some or none of its profits to shareholders. The proposition showed that interest expense reduces tax of the firm, therefore, the value of debt increased in order to maximize the earnings of the firm. Overall, the theory assumed that cost of debt was stable regardless how much the debt was financed. The M&M theorem was also known as Capital Structure Irrelevance Principle.

3.1.2 *Trade-Off Theory*

This theory discussed the idea of how much debt and equity finance in a company was going to be used by balancing its costs and benefits, with the intention of maximizing the firm's value. The theory mainly dealt with cost of financial distress and agency costs. In this theory, tradeoff of costs and benefits of borrowing usually determined the firm's optimal debt ratio. It developed in contradiction of balancing value of interest tax shields and various costs of bankruptcy of financial embarrassment. (Capital Structure Puzzle by Myers, 1984) With the help of determining the cost of financial distress and agency costs, this theory developed to optimize the debt level of a firm in order to determine how much debt and equity to be used for financing.

Cost of Financial Distress, also known as bankruptcy costs, referred directly to the cost of insolvency of a company. This included the legal and administrative costs of bankruptcy; also as agency costs, moral hazard, and contracting costs that could erode the value of the firm even when formal default was avoided. In financial behavior, bull or

risky firms would borrow less when expected costs of distress offset the tax advantage of borrowing. While on the other hand, bear or risk averse firms prefer to borrow more. Due to that, costs of financial distress were caused by actual or estimate defaults. In addition, firms, having tangible assets, would borrow less than the firms that have intangible assets or valuable growth opportunities. Thus, the cost of financial distress was subject to the probability of value lost if trouble does occur. Intangible assets would be included to have value loss when cost of financial distress occurred.

3.1.3 *Static Theory*

The theory determined the optimal capital structure based on the tradeoff of the tax shields and the bankruptcy costs of debt. This stated that there exists a level of debt that a firm could borrow where tax benefit offsets the costs of bankruptcy, also known as the costs of financial distress. This level of debt along the given level of equity was the optimal capital structure.

3.1.4 *Pecking Order Theory*

This theory was developed by Stewart Myers and Nicolas Maljuf in 1984. It had also revised by Lucas and McDonald in 1990. It suggested the order of preference for the firm to fund its capital sources. Based on Myers and Maljuf, the firm would seek first for internal funding because it was the cheapest and does not require private information release which used its profits to finance new projects. As the internal funds were exhausted, firms would then use external financing, such as debt and equity. Firms preferred to finance debt first because it was cheaper than equity financing. Equity

financing would be the last resort to finance new investments. According to Um (2001 as cited in Buferna et al, 2005), emerging companies would most likely choose debt rather than equity from the time when there was a growing pressure to invest in different business opportunities at which it exceeded retained earnings. But Bevan and Danbolt (2002, as cited in Buferna et al, 2005) argued about profitable firms holding less debt since possessing such levels of profits involve increased supply of internal funds.

Based on Kaya (2007), this theory explained 3 things: (1) the inverse relationship of debt ratios and profitability, (2) the behavior of managers to only issue equity when they had no choice or if the firm's shares were overvalued and why markets responded to this negatively; (3) and the reason behind the decision of managers to hold more cash and issued less debt than choosing the trade-off theory.

3.1.5 *Agency Cost Theory*

This theory came about the conflict of two parties, principals (shareholders) and agents (company's executives). As agency cost was high, it implied that agents had high capability in taking their actions such as issuing bonds to the debt holders. Therefore, having higher agency cost resulted to a higher leverage in the company. On the other hand, the shareholders could limit the capability of the agents in order to lessen the agency cost because there would be less conflict between agents and shareholders on making decisions. Thus, there would be less debt of the firm as a result to lower leverage of the firm. To be able to maximize the value of the firm, agency cost should be

minimized. According to Zhang and Li (2008), it was implied that high leverage could be a signal of the existence of high agency costs for the firm.

According to Buferna et al (2005), investment in risky businesses was still one of the management's strategies since this strategy would provide higher returns to its shareholders. Creditors would take the credit when investment fails due to the limitation of liability of the shareholders. (Jensen and Meckling, 1976; cited in Frank and Goyal, 2005) This was what was known as risk shifting. Basic firm characteristics could determine the relationship between the costs and benefits of debt. These characteristics were: Firm Size, Tangibility, Profitability, and Market-to-book ratios. (Rajan and Zingales, 1995)

3.2 Conceptual Framework

3.2.1 *Cost of Capital*

Cost of Capital was used in the field of financial investment that referred to both debt and equity which were the cost of the company's funds. It was also known as the shareholder's required rate of return on a portfolio of the company's existing securities, on an investor's point of view. Cost of capital could be a combination of cost of debt and cost of equity.

Cost of debt constituted the rate of the interest paid. This was used as the interest rate the company applied in paying all of its debt like loans and bonds. As for cost of

equity, it could be defined as the risk-weighted projected return required by investors where the return was unidentified. The cost of equity was stated as the required return of stockholders in a company. There were many ways in calculating cost of equity. Dividend Growth Model and SML Approach were two methods that determine the cost of equity. The Dividend Growth Model was simple to use but was limited to companies that pay dividends at a constant rate. SML approach depended on the required rate of return, market risk premium, and the systematic risk of the asset. The disadvantage of this approach was the necessity of beta coefficient and market risk premium. It also relied on the past data to predict the future.

As both cost of debt and cost of equity were determined, WACC could be calculated in order to use it as a discount rate for a project's projected cash flows. Cost of Capital could also be known as hurdle rate, and cutoff rate.

3.2.2 *Valuation of the Firm*

Valuation was the process of estimating the market value of a financial asset or liability. In assets, valuation could be done on stocks, options, intangible assets, and other investment in marketable securities. In liabilities, valuation could be done on issued bonds in a company. According to the book of Reilly and Brown (2004), there were two approaches in valuation process: Top-down and Bottom-up. The difference between the two approaches was the perceived importance of economic and industry influence on individual firms and stocks. The theory of valuation stated that the value of an asset was the present value of its expected returns. It required the estimation of stream of expected

returns and the required rate of return on investment. There were two techniques in estimating firm's valuation: Discounted Cash Flow Techniques and Relative Valuation Techniques. In Discounted Cash Flow Techniques, it could be used when there was dividend, operating cash flow, and free cash flow to equity. It was dependent on growth rates and discount rates. As for Relative Valuation Techniques, it was reflected from the market perceptions and moods than discounted cash flow valuation. Generally, relative valuation required less information than discounted cash flows. There were also other cases that relative valuation could find weak points of the discounted cash flow valuations and probably fix the problems. Important information from Damodaran was that most of the people in the Wall Street used relative valuation than discounted cash flow valuation.

However, the study would be using a different method. First, the study would calculate the market value of debt, average interest over cost of debt, and market value of equity, average earnings available to shareholders over cost of equity. After that, the researchers would sum the market value of debt and market value of equity in order to acquire the total market value of a company.

3.2.3 *Price Earnings Ratio*

Price Earnings Ratio, also known as earnings multiple, was one of the most important numbers available when evaluating stock price because this measured how expensive the stock was. Benjamin Graham, Father of Value Investing, preached this ratio as one of the fastest and most accurate ways of determining the aggregate amount of

stock. (Kennon, n.d.) According to Reilly and Brown (2004), Price Earnings Ratio values the stock based on expected annual earnings. Thus, P/E ratio was determined by expected dividend payout ratio, required rate of return on the stock, and expected growth rate of dividends. A small change of required rate of return or expected growth rate would have a large impact on the earnings multiplier. (Reilly and Brown, 2004) There were several variations of P/E ratio. One of them was the Trailing P/E ratio. This was presented in the newspaper and other stock tables. The other variation was the Forward P/E ratio. This estimated ratio was used by Wall Street.

3.2.4 Financial Leverage

This ratio defines the proportion of the debt and equity of a firm. It gives the idea of how the company financed its debt and equities. Generally, this measures the company's ability to meet its long term obligations. There were several methods on how to compute the financial leverage. Such methods would be asset over equity ratio, debt to asset ratio, debt to equity ratio, and interest coverage. The main factors of computing leverage ratio were total debt, total equity, total assets, and interest expenses. In our regression model, we used the asset over equity ratio.

3.2.5 Firm Size

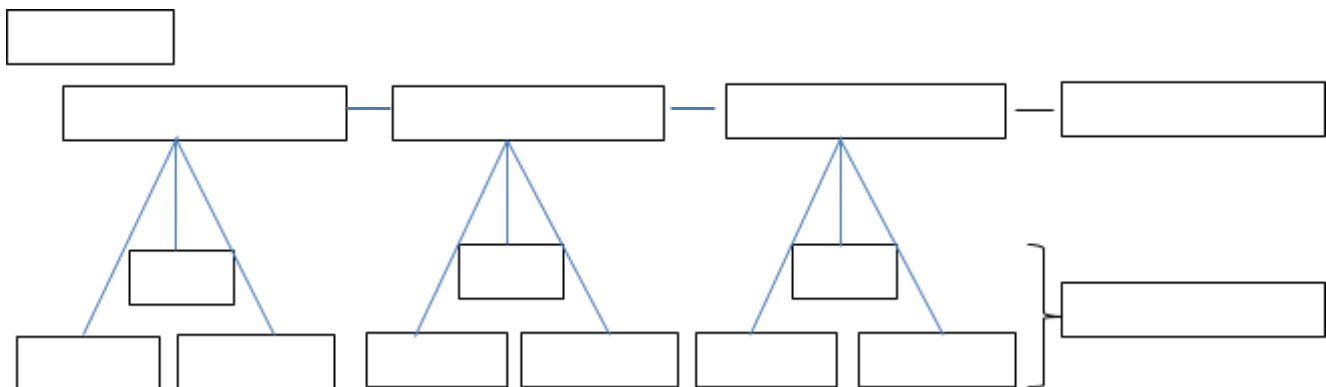
Firm size was used as a method in categorizing companies according to their size for government purposes. This was also a way of comparing other companies in the industry. Firm size could be determined by sales, total assets, and total equities. We used

the logarithm of sales as a proxy variable for firm size in our regression model. It was also one of our dependent variables.

3.2.6 Beta

Beta was the measurement of the systematic risk of a security that compares with the market as a whole. It also measured the volatility of a stock in the market. In cost of capital, beta was used in the capital asset pricing model (CAPM). Theoretically, the higher the beta implied more volatility of a security. It was also called as the beta coefficient.

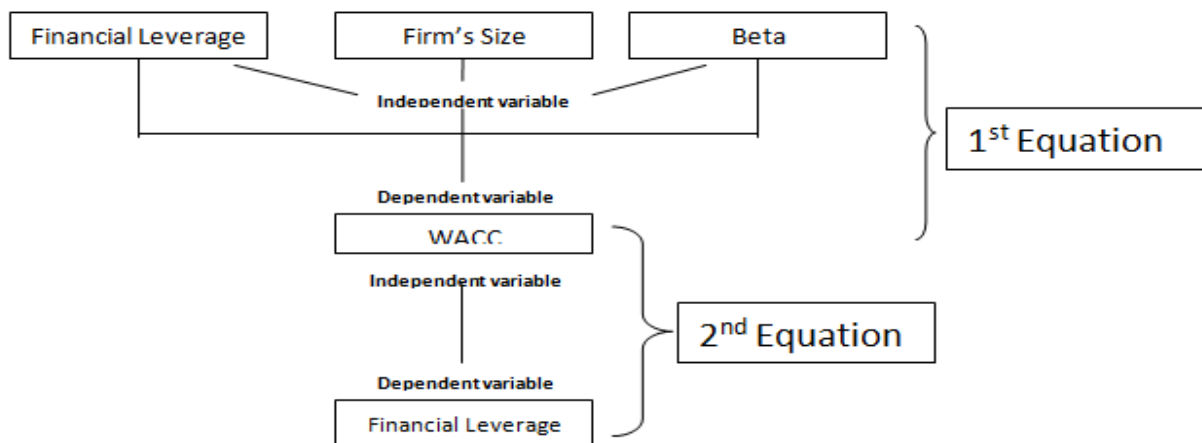
3.3 Operational Framework



The diagram above illustrated the relationship between the variables and how to investigate this study. The independent variable was the financial leverage. On the other hand, the dependent variables were cost of capital, price earnings ratio, and valuation of the firm. The first model divided the sampled companies into two groups, Low-levered and High-levered, for the analysis of financial leverage on cost of capital. The low levered companies contained the

number of companies having the degree of financial leverage lower than the median value. On the other hand, the high-levered companies contained the number of companies having the degree of financial leverage higher than the median value. In M&M theorem, either the financial leverage was levered or unlevered; there was no impact on the cost of capital. However, it was believed in a traditional approach that increasing financial leverage at an early stage of the business resulted to a decrease of cost of capital. After a certain degree level of financial leverage, the financial leverage would increase faster than the cost of capital. Therefore, the relationship of low degree of financial leverage and cost of capital was to be in a negative effect. While, the relationship of high degree of financial leverage and cost of capital was to be in positive effect. Concerning to Price Earnings Ratio, the second model related the capital structure of the firm and the price earnings ratio. Borrowing money from banks and other financial intermediaries included interest expense; that was why having more debt results to more interest expense. When debt was less than the income generated from the firm; the remaining profit would lead to the shareholders' equity thereby earning their share without investing money to the company. To associate financial leverage and Price Earnings Ratio, there would be a negative impact with each other because as debt decreases, there was an increase in net profit and Earnings per Share (EPS), market price of shares. As for the effect of financial leverage, there would be a negative effect when debt decreases. The third model related the impact of financial leverage on valuation of the firm. Theoretically, it was believed that the issuance of debt increases the value of firm, causing a levered firm. After testing these three models by using Karl Pearson's Coefficient of Correlation, the study would analyze the relationship between the financial leverage, cost of capital, price earnings ratio, and valuation of the firm. To test this analysis, the study used the correlation and multiple correlations between the variables.

Figure 3.2



In figure 3.2 above, this diagram illustrated the different variables to be applied in our regression model. We would be using the two stage least squares (2sls) model to determine the optimal debt level. To define the two-stage least squares model, it was basically a method that involves two applications of ordinary least squares. Therefore, in order to account for the interdependence of the two endogenous variables, cost of capital and financial leverage, the researchers used a simultaneous equations model estimated by the two stage least squares method. Thus, we formulated 2sls model to capture the researchers' assumptions. In the first equation, the dependent variable was the weighted average cost of capital (WACC); on the other hand, the determinants of WACC to be used were financial leverage, size of the firm, and beta coefficient. In relations on cost of capital and financial leverage, the cost of capital had an effect on the level of leverage. If the firm's leverage was increased, the debt of the firm increased that result to an increase of WACC because the risk of the firm followed with the debt. For the size of the firm, this related the cost of capital because if the size of the firm was large, it had lower interest rate on borrowing funds because it had higher credit ratings. Thus, this would lead to higher leverage and decreases cost of capital. Concerning on the beta coefficient to cost of

capital, beta was the volatility of the return of the stocks but having higher beta would also result to higher risk. Hence, an increase in beta would lead to an increase in cost of capital. After determining the cost of capital (WACC), the second equation would be formulated. In the second equation, the dependent variable was the financial leverage; while on the other hand, the independent variable was the cost of capital. High cost of capital would result a decrease in financial leverage because as the company's cost of capital was high, there would be likely no new investments in the company. This showed that there would be less debt in the company.

3.3.1 *Operational Definition of Terms*

1. Beta – systematic risk of a security
2. Cost of capital – was measured by the total debt over the total equity. The study would be using the WACC to estimate this
3. Determinants – Variables to be used in the test
4. Financial Leverage – was the use of money to borrow in order to increase the production volume, sales, and earnings. Measured as the ratio of total debt to total assets
5. Firm size – was measured by using the log of sales.
6. Market Capitalization – was measured by multiplying the outstanding shares of the company to the share price in the last trading day of the year.

7. Price Earnings Ratio – equals to the market value per share over the earnings per share

CHAPTER IV

METHODOLOGY OF THE STUDY

4.1 Research Design

The study used an explanatory design to describe the relationship of firms' cost of capital, valuation of debt and equity, price earning ratio to their financial leverage. Those firms publicly listed in the Philippine Stock Exchange (PSE) under construction, infrastructure and allied services. The study also determined the causality relationship between the variables. This method will help identify whether the independent variables have significant effects to the dependent variable.

Formulas:

- Financial Leverage = total asset / total equity.
- Firm Size = log of revenue
- Asset Turnover Ratio = revenue / total assets
- Beta
- Weighted Average Cost of Capital (WACC) = (net worth / total assets) * K_e + (total external liabilities / total assets) * K_d of capital.

Valuation

- Market Value of Debt = average interest / cost of debt
- Market Value of Equity = average earning available to shareholders / cost of equity
- Total Market Value = Market Value of Debt + Market Value of Equity

Price Earnings Ratio = Price per share / earnings per share

Total valuation = f (FL, Ko, Price Earning Ratio)

4.2 Sampling Design

To achieve a valid sample, it must be accurate and precise. For a sample to be valid, accuracy must not be biased from the sample and precise in representing the population. The focus of the study was in the construction, infrastructure and allied services sector of publicly listed firms in the PSE. These firms were assumed to be accurate and precise as they are publicly listed and are mandated and closely watched to disclose. It is assumed that they reflect marked-to-market values of their financial performances. Therefore, the chosen sample is a good sample for the population of the whole sector.

The list of firms were as follows: EEI Corporation, Holcim Philippines, Inc., Phinma Corporation, Republic Cement Corporation, Southeast Asia Cement Holdings, Inc., TKC Steel Corporation, and Vulcan Industrial & Mining

There is no definite size for a sample to represent the population. The study will focus on a sector that is agreeable to be a good representation of the population. The study's sample size is seven.

4.3 Data Description and Collection Method

Secondary data sources will be employed in the study to achieve the objectives of the study. The data for the firms' cost of capital, valuation of debt and equity, and price earning ratio will be collected from the PSE and SEC.

Selected companies based on availability of data under the construction, infrastructure and allied services sector in the PSE. Seven companies under construction, infrastructure and allied services during the period will be used for the purpose of our study. The end of the year data will be used to represent the annual reports to stay consistent.

Annual reports and computed financial ratios for all sets of data from the years 2000-2009 will be used for the purpose of the study.

4.4 Data Analysis Methods

A panel type of data would be used to allow for the observation of variables over a period of time of the seven companies under the construction, infrastructure and allied services sector. The data set that will be used for the variables are from period to period of each company. Given that the study will use seven different companies to define the

relationship of such variables for a period of time, panel data analysis would be the most appropriate analysis for the study. The regression values our study will look at Karl Pearson's coefficient correlation and t-test.

Karl Pearson's coefficient correlation is a quantity closely related to but conceptually very much different from r^2 , which is a measure of the degree of association between two variables (Gujarati, D., 2004). Ranging +1, perfect correlation of x and y (y increases as x increases), to -1, (y increases as x decreases). A coefficient of correlation of 0 have no linear correlation between x and y.

For the t-test, the study will use a 1%, and 5% significance level to determine the relationship between the variables. After computing, independent variables with t-values greater than 2.00 will be considered significant. The study will use a software called Gretl, or also known as Gnu Regression, Econometrics and Time-Series Library software. Gretl will be used in a multiple regression analysis to verify the effects of cost of capital, market value of debt and equity, price earning ratio to the firm's financial leverage.

4.4.1 Test of Assumptions

Preliminary tests for multicollinearity, heteroscedasticity, and autocorrelation would be done to investigate whether the assumption of the classical linear regression model (CLRM) would hold true to indicate the use of the correct model.

One of the assumptions of the CLRM is that there is no exact linear relationship among explanatory or independent variables (Gujarati, D., 2004). This means that the model should have no perfect multicollinearity. In the case of perfect multicollinearity, the individual regression coefficients or their standard errors can no longer be estimated. In cases of high multicollinearity, the marginal or individual contribution of the variable whose t value is low cannot be assessed. The variance inflation factor (VIF) can be used as a detection measure of multicollinearity in the model. VIF provides an index that measures how much the variance of an estimated regression coefficient is increased by collinearity. A cut off of VIF greater than 10 would indicate the presence of multicollinearity in the model.

Another important assumption of the CLRM is the variance of each disturbance term must be constant across all observations. If this is not the case, a situation of heteroscedasticity or unequal variance is present in the model (Gujarati, D., 2004). Once heteroscedasticity is present in the model, OLS estimators are no longer efficient because the variance is not constant. For the purpose of the study, heteroscedasticity will be tested using the White, and Breusch-Godfrey test. If β is significant in the revised model of White test is significant, then heteroscedasticity is present. If chi-square is significant in Breusch-Godfrey test then, heteroscedasticity is also present.

4.5 Methodological Limitations

Sampling was biased, too small, and may not be valid and reliable. The data collection was reliant on the source of the financial figures given by BLOOMBERG. Another limitations the study would encountered was the complexity of fixing (remedy or doing nothing) the model if it has multicollinearity, or heteroscedasticity.

CHAPTER V

ANALYSIS AND INTERPRETATION OF RESULTS

5.1 Karl Peason's Coefficient Correlation

Using Karl Pearson's coefficient correlation, we first correlated financial leverage to market capitalization; then correlated it to P/E ratio. Lastly, we correlated financial leverage to WACC. After that, we separated the low levered and high levered companies by using correlation matrix in Gretl. We then separately correlated the low levered and high levered companies to their market capitalization, P/E ratio, and WACC. After testing it, we use two stage least squares (2sls) determining the optimal debt level. In testing heteroscedasticity and mutlicollinearity between variables, we use the OLS model. All data can be found in Appendix B

5.1.1 Industry Coefficient Correlation

Correlation coefficients, using the observations 1:1 - 7:9
5% critical value (two-tailed) = 0.2480 for n = 63

mkt cap	fin lev	
1.0000	-0.1824	mkt cap
	1.0000	fin lev

$\text{corr}(\text{fin_lev}, \text{mkt_cap}) = -0.18238845$

Under the null hypothesis of no correlation:

$t(61) = -1.4488$, with two-tailed p-value 0.1525

Correlation coefficients, using the observations 1:1 - 7:9
5% critical value (two-tailed) = 0.2480 for $n = 63$

pe rat	fin lev	
1.0000	0.0068	pe rat
	1.0000	fin lev

$\text{corr}(\text{fin_lev}, \text{pe_rat}) = 0.00676871$

Under the null hypothesis of no correlation:

$t(61) = 0.0528665$, with two-tailed p-value 0.9580

Correlation coefficients, using the observations 1:1 - 7:9
5% critical value (two-tailed) = 0.2480 for $n = 63$

Wacc	fin lev	
1.0000	-0.1758	Wacc
	1.0000	fin lev

$\text{corr}(\text{fin_lev}, \text{wacc}) = -0.17579844$

Under the null hypothesis of no correlation:

$t(61) = -1.39475$, with two-tailed p-value 0.1681

The result for pairing financial leverage with market cap, pe ratio and wacc were insignificant in the industry level.

Table 5.1a

Fin lev	average	industry 2.43
Cmt	1.50	low
Eei	3.22	high
hlcm	1.68	low
phn	3.65	high
rcm	2.84	high
t	2.56	high
vul	1.55	low

Table 5.1b

Fin lev - median		
industry	1.97	LEVERED
cmt	1.15	LOW
eei	3.13	HIGH
hlcm	1.64	LOW
phn	1.47	LOW
rcm	2.84	HIGH
t	2.15	HIGH
vul	1.55	LOW

The table 5.1a shows the average of a firm for the period of 2001-2009. The industry was computed by averaging all of the firms' financial leverage during 2001-2009. Table 5.1b on the other hand shows the median of a firm for the same period.g

5.1.2 High Levered Firms Coefficient Correlation

INDUSTRY AVERAGE

HIGH LEVERED

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for n = 36

fin lev	wacc	
1.0000	-0.0526	fin lev
	1.0000	wacc

$\text{corr}(\text{fin_lev}, \text{wacc}) = -0.05257316$

Under the null hypothesis of no correlation:

$t(34) = -0.306976$, with two-tailed p-value 0.7607

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for $n = 36$

fin lev	pe rat	
1.0000	-0.1358	fin lev
	1.0000	pe rat

$\text{corr}(\text{fin_lev}, \text{pe_rat}) = -0.13583840$

Under the null hypothesis of no correlation:

$t(34) = -0.799478$, with two-tailed p-value 0.4296

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for $n = 36$

fin lev	mkt cap	
1.0000	-0.1859	fin lev
	1.0000	mkt_cap

$\text{corr}(\text{fin_lev}, \text{mkt_cap}) = -0.18594800$

Under the null hypothesis of no correlation:

$t(34) = -1.1035$, with two-tailed p-value 0.2776

The result for pairing financial leverage with market cap, pe ratio and wacc were insignificant for high levered firms.

INDUSTRY MEDIAN

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for n = 27

fin_lev	mkt_cap	
1.0000	-0.2856	fin_lev
	1.0000	mkt_cap

$\text{corr}(\text{fin_lev}, \text{mkt_cap}) = -0.28563568$

Under the null hypothesis of no correlation:

$t(25) = -1.49027$, with two-tailed p-value 0.1487

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for n = 27

fin_lev	pe_rat	
1.0000	-0.0355	fin_lev
	1.0000	pe_rat

$\text{corr}(\text{fin_lev}, \text{pe_rat}) = -0.03545433$

Under the null hypothesis of no correlation:

$t(25) = -0.177383$, with two-tailed p-value 0.8606

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for n = 27

fin_lev	wacc	
1.0000	-0.3836	fin_lev
	1.0000	wacc

$\text{corr}(\text{fin_lev}, \text{wacc}) = -0.38356609$

Under the null hypothesis of no correlation:

$t(25) = -2.07667$, with two-tailed p-value 0.0483

5.1.3 Low Levered Firms Coefficient Correlation

INDUSTRY AVERAGE

LOW LEVERED

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for $n = 27$

fin_lev	wacc	
1.0000	-0.1169	fin_lev
	1.0000	wacc

$\text{corr}(\text{fin_lev}, \text{wacc}) = -0.11688313$

Under the null hypothesis of no correlation:

$t(25) = -0.588449$, with two-tailed p-value 0.5615

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for $n = 27$

fin_lev	pe_rat	
1.0000	-0.1332	fin_lev

	1.0000	pe_rat
--	--------	--------

$\text{corr}(\text{fin_lev}, \text{pe_rat}) = -0.13324765$

Under the null hypothesis of no correlation:

$t(25) = -0.672233$, with two-tailed p-value 0.5076

Correlation coefficients, using the observations 1:1 - 3:9
5% critical value (two-tailed) = 0.3809 for $n = 27$

fin_lev	mkt_cap	
1.0000	-0.0600	fin_lev
	1.0000	mkt_cap

$\text{corr}(\text{fin_lev}, \text{mkt_cap}) = -0.05995677$

Under the null hypothesis of no correlation:

$t(25) = -0.300324$, with two-tailed p-value 0.7664

The result for pairing financial leverage with market cap, pe ratio and wacc were insignificant for low levered firms.

INDUSTRY MEDIAN

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for $n = 36$

fin_lev	mkt_cap	
1.0000	-0.1348	fin_lev
	1.0000	mkt_cap

$\text{corr}(\text{fin_lev}, \text{mkt_cap}) = -0.13483948$

Under the null hypothesis of no correlation:

$t(34) = -0.793489$, with two-tailed p-value 0.4330

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for $n = 36$

fin_lev	pe_rat	
1.0000	-0.0415	fin_lev
	1.0000	pe_rat

$\text{corr}(\text{fin_lev}, \text{pe_rat}) = -0.04153194$

Under the null hypothesis of no correlation:

$t(34) = -0.24238$, with two-tailed p-value 0.8099

Correlation coefficients, using the observations 1:1 - 4:9
5% critical value (two-tailed) = 0.3291 for $n = 36$

fin_lev	wacc	
1.0000	0.0056	fin_lev
	1.0000	wacc

$\text{corr}(\text{fin_lev}, \text{wacc}) = 0.00564969$

Under the null hypothesis of no correlation:

$t(34) = 0.0329436$, with two-tailed p-value 0.9739

All correlation matrix results have given p-value of > 0.05 . all of them are insignificant whether they are taken as a whole industry, or separated by the level of financial leverage.. There is no impact on financial leverage with cost of capital, market capitalization and price earning

ratio. Therefore, all null hypotheses are accepted. The results are consistent with the results of Bhayani's earlier study.

- Financial Leverage (-): We expect a negative relationship between financial leverage and cost of capital
- Firm Size (+/-): Log(Revenue) was used as a proxy for firm size. Firm size is positively related to the weighted average cost of capital assuming large firms have a greater access to capital markets, therefore leading to lower levels of leverage and increases overall cost of capital. Firm size may also be negatively related to cost of capital. Since large firms have lower bankruptcy risks, leading to higher levels of leverage and decreases overall cost of capital.
- Beta(+): Beta is expected to have a positive relationship to cost of capital. Since a high beta would lead to low leverage while increasing overall cost of capital

5.2 Regression Analysis

5.2.1 Two-stage Least Squares

Table 5.2

Instrumental variables (2SLS) regression

Source	SS	df	MS	Number of obs = 63		
Model	-.539064448	4	-.134766112	F(4, 58) = 2.03		
Residual	.626635876	58	.010804067	Prob > F = 0.1026		
Total	.087571428	62	.001412442	R-squared = .		
				Adj R-squared = .		
				Root MSE = .10394		

wacc	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
finlev	-.2480918	.0873406	-2.84	0.006	-.4229231	-.0732605
finlev2	.026867	.0096002	2.80	0.007	.0076501	.0460839

lnrev		.0128323	.0082657	1.55	0.126	-.0037133	.029378
beta		-.0182334	.0221614	-0.82	0.414	-.0625944	.0261275
_cons		.2084395	.1544133	1.35	0.182	-.1006522	.5175313

Instrumented:	finlev						
Instruments:	finlev2 lnrev beta wacc						

$$Wacc_{i,t} = a + \beta_1 \text{fin_lev} + \beta_2 \text{fin_lev_2} + \beta_3 \text{l_revenue} + \beta_4 \text{beta}$$

$$\text{fin_lev} = a + \beta_1 Wacc_{i,t}$$

Table 5.1 shows the regression results of the above regression model on STATA. In order to account for the joint determination of the degree of financial leverage and cost of capital, a two-stage least squares regression was used. Financial leverage is statistically significant in affecting cost of capital as $P > |t|$ which is 0.006 is less than 0.05. For every 1 unit increase in financial leverage, cost of capital decreases by 0.2480918. The result from the 2SLS regression is the same in our a-priori expectations, that financial leverage negatively affect WACC. Due to this effect, companies in this sub-sector should use more debt to finance new projects so that they can reach their optimal debt level. The finlev^2 variable is statistically significant because its $P > |t|$ which is 0.007 is less than 0.05. Thus, it can be used to get the optimal debt level by using first and second condition order. The variable $\ln(\text{rev})$, a proxy for firm size, is statistically insignificant because its $P > |t|$ which is 0.126 is not less than 0.10. Also, the beta coefficient is statistically insignificant based on the regression table above.

Using the regression equation for deriving the optimal debt level:

$$Wacc_{i,t} = 0.2084395 - 0.2480918(\text{fin_lev}) + 0.026867(\text{fin_lev_2}) + 0.01283323(\text{l_revenue}) - 0.0182334(\text{beta})$$

Getting the first – order and second – order conditions for a minimum:

$$(\text{partial derivative of wacc w/ respect to the partial derivative financial leverage}) = -0.2480918 + (0.053734)(\text{fin_lev}) = 0$$

$$(\text{second partial derivative of wacc w/ respect to the partial derivative financial leverage}) < 0$$

After deriving the regression equation, the critical value of financial leverage would be 3.617035769. This value is the optimal debt ratio for the construction, infrastructure and allied services sub-sector of the PSE.

Table 5.3

Difference of D/E leverage to optimal leverage			
	D/E	Optimal debt level	Difference
Cmt	1.1600	3.617	-2.4570
Eei	1.5100	3.617	-2.1070
Hlcm	0.3900	3.617	-3.2270
Phn	0.3700	3.617	-3.2470
Rcm	0.5600	3.617	-3.0570
T	1.2600	3.617	-2.3570
Vul	0.6900	3.617	-2.9270

Table 5.2 above shows the difference between the computed optimal capital structure of the sub-sector to the actual capital structure of the firms for the year 2009. The firms included in the study can be classified as still being low-levered. This aversion in debt may be caused by the fear of bankruptcy costs and static state of the industry that the firms operate in. However, the

outlook for the construction industry is positive. Which may indicate that the firms included in the study may be more willing to increase their debt ratio to further minimize their cost of capital. This is also implies why the relationship between financial leverage and WACC is negatively effect in the regression table.

5.2.2 Ordinary Least Squares Method

Model 12: Pooled OLS, using 63 observations

Included 7 cross-sectional units

Time-series length = 9

Dependent variable: wacc

	coefficient	std. error	t-ratio	p-value	

const	0.0905416	0.0434111	2.086	0.0417	**
l_revenue	0.00180284	0.00222927	0.8087	0.4222	
fin_lev	-0.0170925	0.00867615	-1.970	0.0539	*
fin_lev__2	0.00172614	0.000957317	1.803	0.0769	*
asset_turn	-0.0160653	0.00240008	-6.694	1.20e-08	***
pe_rat	-6.04725e-05	8.62642e-05	-0.7010	0.4862	
mkt_cap	-6.02543e-07	4.02699e-07	-1.496	0.1403	
beta	-0.00951100	0.00566452	-1.679	0.0988	*
Mean dependent var	0.078443	S.D. dependent var	0.037258		
Sum squared resid	0.038932	S.E. of regression	0.026605		
R-squared	0.547649	Adjusted R-squared	0.490077		
F(7, 55)	9.512417	P-value(F)	1.11e-07		
Log-likelihood	143.3630	Akaike criterion	-270.7260		
Schwarz criterion	-253.5810	Hannan-Quinn	-263.9828		

rho 0.104174 Durbin-Watson 1.546647

Excluding the constant, p-value was highest for variable 5 (pe_rat)

$$Wacc_{i,t} = a + \beta_1 l_revenue + \beta_2 fin_lev + \beta_3 fin_lev_2 + \beta_4 asset_turn + \beta_5 pe_rat + \beta_6 mkt_cap + \beta_7 beta$$

5.3 Test for Heteroskedasticity and Multicollinearity

5.3.1 Heteroskedasticity Test

Table 5.4

White's test for heteroskedasticity
 OLS, using 63 observations
 Dependent variable: uhat^2
 Omitted due to exact collinearity: sq_fin_lev

	coefficient	std. error	t-ratio	p-value	
const	-0.0481657	0.0408999	-1.178	0.2489	
l_revenue	0.00380888	0.00345362	1.103	0.2795	
fin_lev	0.0233582	0.0197208	1.184	0.2462	
fin_lev__2	-0.00444864	0.00357161	-1.246	0.2232	
asset_turn	0.00914366	0.00753978	1.213	0.2354	
pe_rat	3.81115e-05	0.000142381	0.2677	0.7909	
mkt_cap	-1.64788e-06	9.39698e-07	-1.754	0.0904	*
beta	0.000697798	0.00489389	0.1426	0.8876	
sq_l_revenue	-6.52232e-05	8.40393e-05	-0.7761	0.4442	
X2_X3	-0.00134124	0.00108485	-1.236	0.2266	
X2_X4	0.000335323	0.000205534	1.631	0.1140	
X2_X5	-0.000480264	0.000236484	-2.031	0.0519	*
X2_X6	-1.99311e-06	5.69940e-06	-0.3497	0.7292	
X2_X7	4.60995e-08	4.67447e-08	0.9862	0.3325	
X2_X8	0.000254491	0.000222946	1.141	0.2633	
X3_X4	-0.000499022	0.000381300	-1.309	0.2013	
X3_X5	0.000616382	0.00427009	0.1443	0.8863	
X3_X6	2.09965e-05	8.97294e-05	0.2340	0.8167	
X3_X7	6.69237e-07	3.72833e-07	1.795	0.0835	*
X3_X8	-0.00429095	0.00221297	-1.939	0.0626	*
sq_fin_lev__	2.38598e-05	1.98926e-05	1.199	0.2404	
X4_X5	-1.21855e-05	0.000795116	-0.01533	0.9879	
X4_X6	-5.46294e-06	1.61288e-05	-0.3387	0.7374	
X4_X7	-1.50352e-07	8.52076e-08	-1.765	0.0885	*
X4_X8	0.000648565	0.000381232	1.701	0.1000	*
sq_asset_turn	-0.000121320	0.000156844	-0.7735	0.4457	
X5_X6	2.35538e-05	2.06532e-05	1.140	0.2638	

X5_X7	-6.11611e-08	1.68624e-07	-0.3627	0.7195
X5_X8	0.000679371	0.000849701	0.7995	0.4307
sq_pe_rat	2.79607e-08	5.20580e-08	0.5371	0.5954
X6_X7	-3.12286e-09	2.51655e-09	-1.241	0.2249
X6_X8	-2.06795e-05	5.12594e-05	-0.4034	0.6897
sq_mkt_cap	1.32469e-12	2.50733e-12	0.5283	0.6014
X7_X8	-8.39479e-08	7.12627e-08	-1.178	0.2487
sq_beta	0.000139194	0.000336410	0.4138	0.6822

Warning: data matrix close to singularity!

Unadjusted R-squared = 0.781409

Test statistic: $TR^2 = 49.228788$,
with p-value = $P(\text{Chi-square}(34) > 49.228788) = 0.044136$

5.3.2 Multicollinearity Test

Variance Inflation Factors

Minimum possible value = 1.0

Values > 10.0 may indicate a collinearity problem

l_revenue	1.737
fin_lev	13.385
fin_lev__2	11.702
asset_turn	1.212
pe_rat	1.024
mkt_cap	1.559
beta	1.105

$VIF(j) = 1/(1 - R(j)^2)$, where $R(j)$ is the multiple correlation coefficient between variable j and the other independent variables

Properties of matrix $X'X$:

1-norm = 8.9444203e+09
Determinant = 1.8487829e+27
Reciprocal condition number = 3.8035128e-11

Using OLS method, we found out that there is a collinearity problem in financial leverage indicating that its value in VIF was more than 10. Still, the VIF value was acceptable because we really intended to put a squared value of financial leverage to get the optimal level. In testing heteroscedasticity, we used White's Test and Breusch-Pagan Test. Looking at its p-value, there is no heteroscedasticity problem because its value is less than 0.05.

CHAPTER VI

CONCLUSIONS AND RECOMMENDATIONS

This chapter includes the conclusion and recommendation of the study. It summarizes all our empirical findings of the study. Results are being connected I our objectives and hypotheses stated in the paper. Also, we are giving recommendations on how to improve in leading to further studies.

6.1 Conclusions

Based on Bhayani (2009), there are no correlation between financial leverage and weighted average cost of capital or price earning ratio or market cap. In our study, all of our null hypotheses were accepted, resulting that all correlation tests were insignificant. Also, the result does not support the Modigliani and Miller approach about high levered companies resulted to higher valuation than the low levered companies.

Auon and Hesmati (2006) showed the relationships among weighted average cost of capital, financial leverage, firm size and firm's systematic risk. The researchers found out that there is significance on defining weighted average cost of capital with the given independent variables. The regression results paved way for the determination of the optimal debt level with minimized cost of capital. Using these sources, the researchers patterned this study on the aforementioned model. However, the researchers were restricted to choose proxy for a variable because it's irrelevance or availability of the data. The model generated from the study was only interested to investigate the optimal debt level during the time period of 2001-2009 of construction, infrastructure and allied services sub-sector in the Philippines. In order to achieve for the optimal debt level, the researchers developed a multiple regression model using the 2SLS

method with the weighted average cost of capital as the dependent variable and financial leverage, firm size and firm's systematic risk as independent variables.

Results are aligned from a-priori expectations wherein financial leverage was expected to have a negative relationship to cost of capital. The firm size or log of revenue showed positive relationship. Lastly, the firm's systematic risk or beta was expected to have a positive relationship. However, only the financial leverage and its squared term were significant; while, the firm size and beta coefficient were insignificant in the study.

Regarding the values of the coefficients, the OLS results suggest that 1 unit change in financial leverage affects the WACC by -0.0170925 with its squared term by 0.00172614. A change in 1 unit in firm size affects the WACC by 0.0180284. 1 unit change in beta gives an impact of -0.009511 to WACC. 1 unit change in asset tangibility gives 10.0160653 effect to WACC. 1 unit change in price earnings ratio affects WACC by -6.04725e-05. 1 unit change in market capitalization impacts WACC by -6.02543e-07. Holding all else constant, the value of WACC is 0.0905416. In order to obtain the optimal capital structure, we used optimization on WACC with respect to financial leverage. In the 2SLS, The optimization resulted with a critical value for leverage of 3.617035769. It implied that the firm's debt should be 3.617 times more than its equity. The researchers then compared the computed optimal capital structure with the actual capital structure of the firms for 2009. The result indicated that the firms should still increase their debt level in order to achieve the optimal capital structure that minimizes its cost of capital.

In conclusion, a highly levered firm incurs a low cost of capital and vis-a-vie with the exception of exceeding the optimal level of debt which would make a highly levered firm incur a

high cost of capital. The researchers suggest that it may be wise for the management to know the optimal level for them to operate at maximum efficiency.

6.2 Recommendations

Despite the getting the results as expected, this study has its limitations. Researchers have recommendations on how future studies will be more accurate. First, future researchers on this field might also select to increase the number of financial variables analyzed. There may be other variables that can best define the sub-sector's financial leverage or firm size. Perhaps some other financial variable might better provide a better relationship for a firm's WACC. Second, this study was plagued with several inactively traded stocks that have restricted our sample size to 7. This resulted to a lower number of observations than desired. Also, the other industry players are new and irrelevant to study because of data availability. Third, the model generated was low on r-squared and high on P-value(F). The results suggested that the power of predictability was weak but its' combination of variables used was good. Future researchers should add and not drop variables. Lastly, the prospective researchers might opt to look for other models as that might define the optimal debt level. From there, they could cross-examine each model and choose the best one.

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